

Compiler Design Lab Programs

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Course Code: CSA1402

33.School management wants to validate DOB of all students. Write a LEX program to implement it.

Code:

```
%{
#include <stdio.h>

%}

%%

(0[1-9]|[12][0-9]|3[01])[-/.](0[1-9]|1[0-2])[-/.](19[0-9]{2}|20[0-9]{2}) {
    printf("Valid DOB: %s\n", yytext);
}

[^\n]+ { printf("Invalid DOB: %s\n", yytext); }

\n ;

%%

int main(){ yylex(); return 0; }

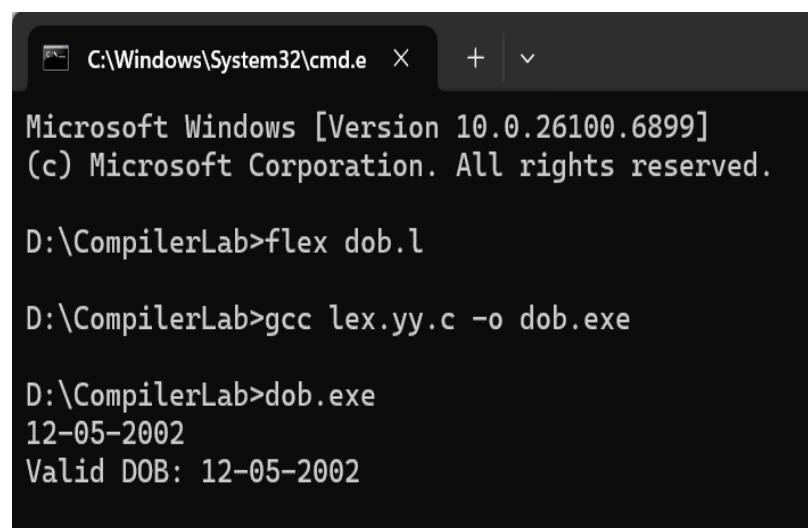
int yywrap(){ return 1; }
```

Output:

12-05-2002

Valid DOB: 12-05-2002

Output Screenshot:



```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex dob.l

D:\CompilerLab>gcc lex.yy.c -o dob.exe

D:\CompilerLab>dob.exe
12-05-2002
Valid DOB: 12-05-2002
```

34. Write a LEX program to check whether the given input is digit or not.

Code:

```
%{  
#include <stdio.h>  
%}  
%%  
[0-9] { printf("It is a Digit: %s\n", yytext); }  
.\n { printf("Not a Digit: %s\n", yytext); }  
%%  
int main(){ yylex(); return 0; }  
int yywrap(){ return 1; }
```

Output:

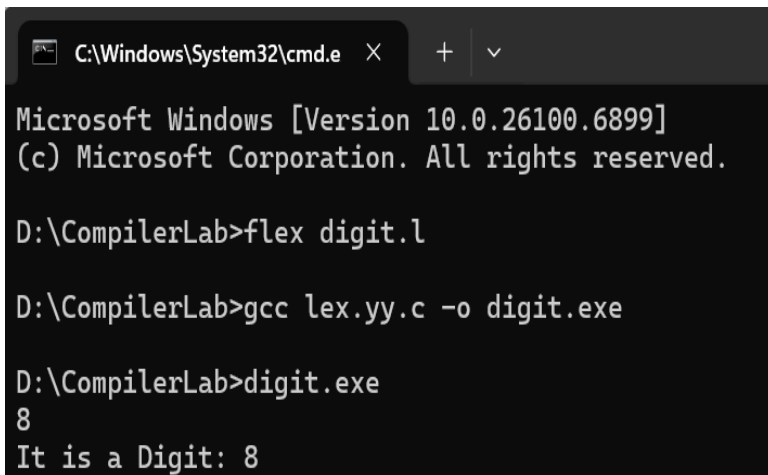
8

a

It is a Digit: 8

Not a Digit: a

Output Screenshot:



```
C:\Windows\System32\cmd.e X + v  
Microsoft Windows [Version 10.0.26100.6899]  
(c) Microsoft Corporation. All rights reserved.  
  
D:\CompilerLab>flex digit.l  
  
D:\CompilerLab>gcc lex.yy.c -o digit.exe  
  
D:\CompilerLab>digit.exe  
8  
It is a Digit: 8
```

35. A School student was asked to do basic mathematical operations. Implement a LEX program to implement the same.

Code:

```
%{  
#include <stdio.h>  
#include <stdlib.h>
```

```

%}
NUMBER [0-9]+
%%

{NUMBER}"+"{NUMBER} { int a,b; sscanf(yytext,"%d+%d",&a,&b); printf("Addition:
%d\n", a+b); }

{NUMBER}"-"{NUMBER} { int a,b; sscanf(yytext,"%d-%d",&a,&b); printf("Subtraction:
%d\n", a-b); }

{NUMBER}"*"{NUMBER} { int a,b; sscanf(yytext,"%d*%d",&a,&b);
printf("Multiplication: %d\n", a*b); }

{NUMBER}"/"{NUMBER} { int a,b; sscanf(yytext,"%d/%d",&a,&b); if(b!=0)
printf("Division: %.2f\n", (float)a/b); else printf("Division by zero\n"); }

[ \t\n]+ ;
. ;
%%

int main(){ yylex(); return 0; }
int yywrap(){ return 1; }

```

Output:

10+20

25-5

6*5

20/4

20/0

Addition: 30

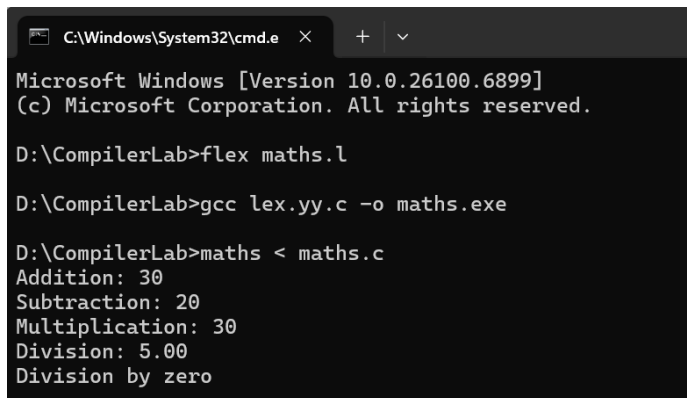
Subtraction: 20

Multiplication: 30

Division: 5.00

Division by zero

Output Screenshot:



```
C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex maths.l

D:\CompilerLab>gcc lex.yy.c -o maths.exe

D:\CompilerLab>maths < maths.c
Addition: 30
Subtraction: 20
Multiplication: 30
Division: 5.00
Division by zero
```

36. Write a LEX program to accept string starting with vowel.

Code:

```
%{
#include <stdio.h>

%}

%%

[AEIOUaeiou][a-zA-Z]* { printf("Valid: %s\n", yytext); }
[a-zA-Z]+ { printf("Invalid: %s\n", yytext); }
.\n ;

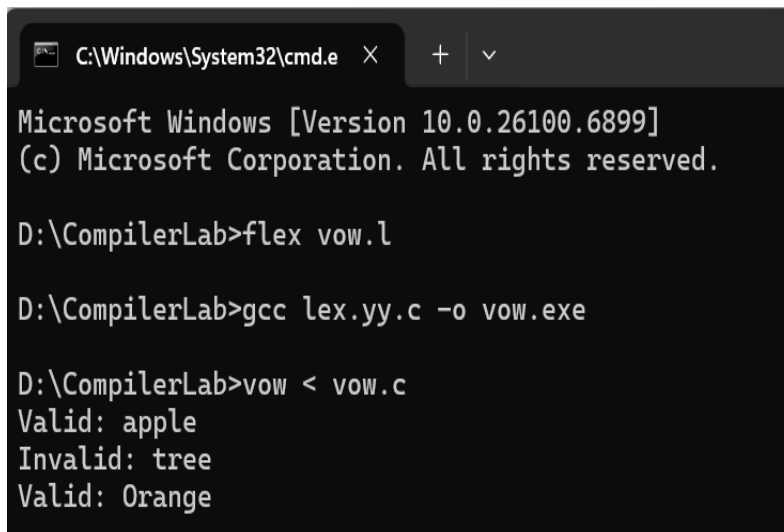
%%

int main(){ yylex(); return 0; }
int yywrap(){ return 1; }
```

Output:

```
apple
tree
Orange
Valid: apple
Invalid: tree
Valid: Orange
```

Output Screenshot:



```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26100.6899]
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D:\CompilerLab>flex vow.l

D:\CompilerLab>gcc lex.yy.c -o vow.exe

D:\CompilerLab>vow < vow.c
Valid: apple
Invalid: tree
Valid: Orange
```